

Sales Analysis

May 1, 2025

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[5]: import sqlite3
import random

conn = sqlite3.connect("sales_data.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS sales (
    id INTEGER PRIMARY KEY,
    product TEXT,
    quantity INTEGER,
    price REAL
)
""")

products = ['Apple', 'Banana', 'Orange', 'Grapes', 'Mango', 'Peach',
            'Pineapple', 'Strawberry', 'Watermelon', 'Blueberry']

sample_data = []
for _ in range(100):
    product = random.choice(products)
    quantity = random.randint(1, 20)
    price = round(random.uniform(1.0, 5.0), 2)
    sample_data.append((product, quantity, price))

cursor.executemany("INSERT INTO sales (product, quantity, price) VALUES (?, ?, ?)", sample_data)

conn.commit()
conn.close()

print("Inserted 100 records into sales_data.db")
```

Inserted 100 records into sales_data.db

```
[6]: import sqlite3
import pandas as pd
import matplotlib.pyplot as plt
```

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conn = sqlite3.connect("sales_data.db")

query = """
SELECT
    product,
    SUM(quantity) AS total_quantity,
    SUM(quantity * price) AS total_revenue
FROM sales
GROUP BY product
"""

df = pd.read_sql_query(query, conn)

print("Sales Summary:")
print(df)

plt.figure(figsize=(10, 6))
df.plot(kind='bar', x='product', y='total_revenue', color='skyblue',
        legend=False)
plt.title("Total Revenue by Product")
plt.ylabel("Revenue")
plt.xlabel("Product")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("sales_chart.png")
plt.show()

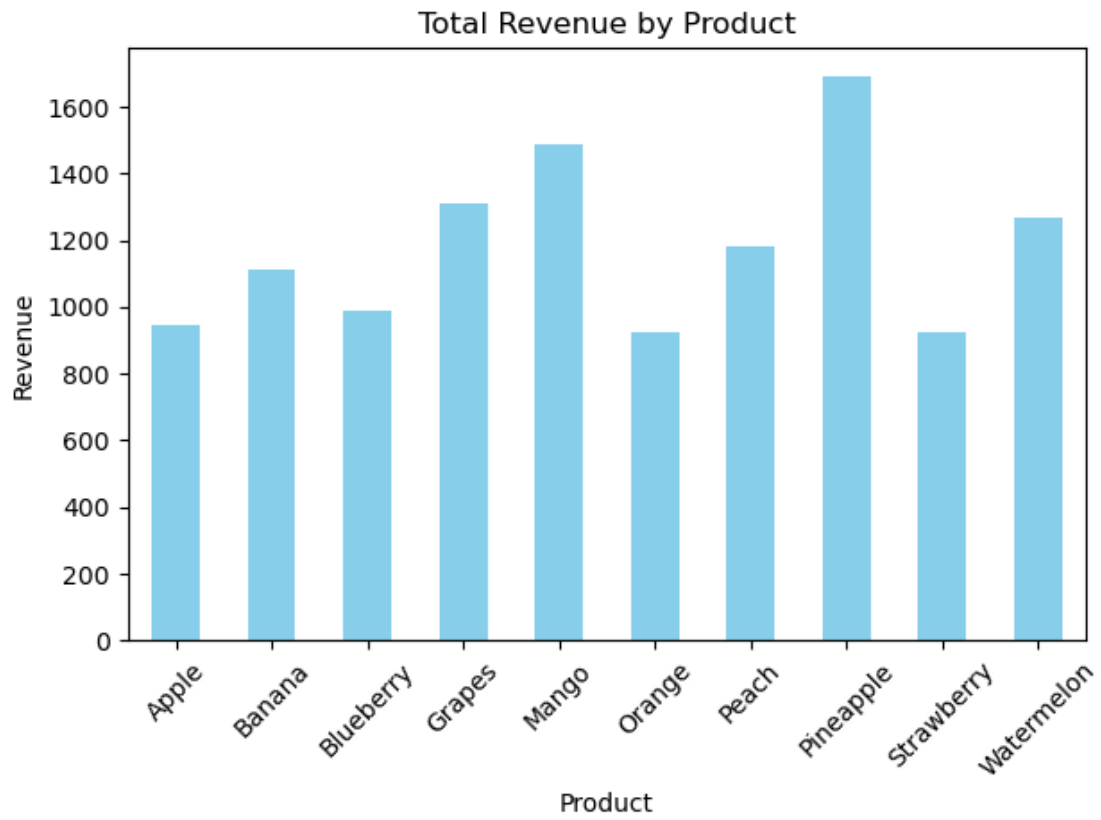
conn.close()

```

Sales Summary:

	product	total_quantity	total_revenue
0	Apple	304	945.18
1	Banana	455	1110.68
2	Blueberry	351	986.91
3	Grapes	487	1308.12
4	Mango	525	1489.55
5	Orange	332	922.64
6	Peach	411	1182.79
7	Pineapple	628	1690.78
8	Strawberry	353	924.80
9	Watermelon	409	1269.62

<Figure size 1000x600 with 0 Axes>



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